

AI-Assisted Coding

Assignment-3.5

Name: K.SUJITH

Batch:45

Question 1: Zero-Shot Prompting (Leap Year Check)

Write a zero-shot prompt to generate a Python function that checks

whether a given year is a leap year.

Task:

- Record the AI-generated code.
- Test with years like 1900, 2000, 2024.
- Identify logical flaws or missing conditions.

The screenshot shows an IDE interface with the following components:

- EXPLORER:** Shows a project structure with 'AI ASSISTED CODING' and files 'assignment_3.5.py' and 'main.py'.
- EDITOR:** Displays the code for 'assignment_3.5.py'. The code defines a function `is_leap_year` that takes a year as input and returns a boolean. It includes a docstring and example usage.
- CHAT:** Shows a prompt: 'Write a Python function named is_leap_year that accepts an integer representing a year. The function should return True if the year is a leap year according to the Gregorian calendar rules (divisible by 4, but not by 100 unless also divisible by 400), and False otherwise. Include type hinting and a docstring.' The response shows the generated code.
- TERMINAL:** Shows the execution of the code. It prompts the user to enter a year and displays the output for 1900 (not a leap year), 2000 (leap year), and 2024 (leap year).

Question 2: One-Shot Prompting (GCD of Two Numbers)

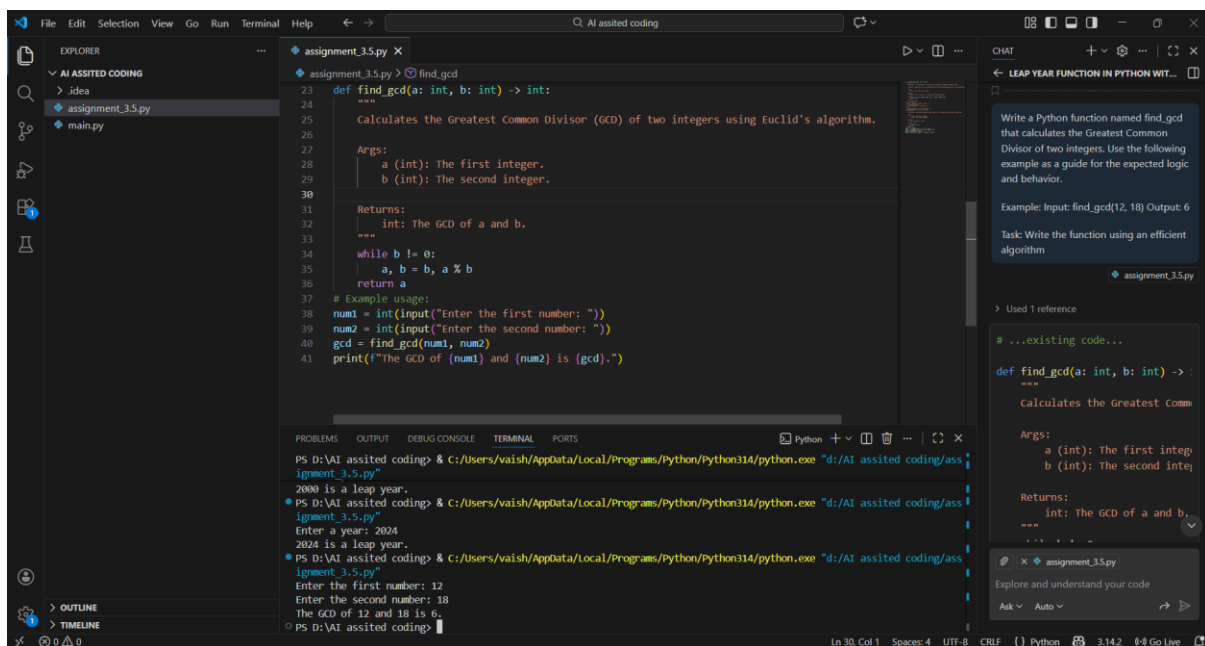
Write a one-shot prompt with one example to generate a Python function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6

Task:

- Compare with a zero-shot solution.
- Analyze algorithm efficiency



The screenshot shows a VS Code editor with a Python file named `assignment_3.5.py`. The code defines a function `find_gcd` that uses Euclid's algorithm to find the GCD of two integers. The function has two arguments, `a` and `b`, and returns the GCD. Below the function definition, there is an example usage section that prompts the user to enter two numbers and prints the result. The terminal output shows the execution of the script, where the user enters 12 and 18, and the output is 6. On the right side, there is a chat window with a prompt that asks to write a Python function named `find_gcd` that calculates the GCD of two integers, using the provided example as a guide. The chat window also shows the function definition and the example usage section.

```
def find_gcd(a: int, b: int) -> int:
    """
    Calculates the Greatest Common Divisor (GCD) of two integers using Euclid's algorithm.

    Args:
        a (int): The first integer.
        b (int): The second integer.

    Returns:
        int: The GCD of a and b.
    """
    while b != 0:
        a, b = b, a % b
    return a

# Example usage:
num1 = int(input("Enter the first number: "))
num2 = int(input("Enter the second number: "))
gcd = find_gcd(num1, num2)
print(f"The GCD of {num1} and {num2} is {gcd}."
```

Write a Python function named `find_gcd` that calculates the Greatest Common Divisor of two integers. Use the following example as a guide for the expected logic and behavior.

Example: Input: `find_gcd(12, 18)` Output: 6

Task: Write the function using an efficient algorithm

```
def find_gcd(a: int, b: int) -> int:
    """
    Calculates the Greatest Common Divisor (GCD) of two integers using Euclid's algorithm.

    Args:
        a (int): The first integer.
        b (int): The second integer.

    Returns:
        int: The GCD of a and b.
    """
    while b != 0:
        a, b = b, a % b
    return a
```

Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python

function that computes the Least Common Multiple (LCM).

Examples:

- Input: 4, 6 → Output: 12
- Input: 5, 10 → Output: 10
- Input: 7, 3 → Output: 21

Task:

- Examine how examples guide formula selection.
- Test edge cases

The screenshot shows a VS Code editor with a Python file named `assignment_3.5.py`. The code defines a function `calculate_lcm(a: int, b: int) -> int:` and includes a main block that takes user input and prints the LCM. The terminal shows the execution of the script with two test cases: (4, 6) resulting in 12, and (5, 10) resulting in 10. The right sidebar shows a chat window with a prompt about LCM calculation and a response that includes the function definition.

```
def calculate_lcm(a: int, b: int) -> int:
    """
    The LCM is the smallest positive integer that is a multiple of both a and b.

    Args:
        a (int): The first integer.
        b (int): The second integer.

    Returns:
        int: The LCM of a and b.
    """
    if a == 0 or b == 0:
        return 0 # LCM is undefined for zero, but per examples, assume positive integers
    gcd = find_gcd(a, b)
    return abs(a * b) // gcd

a = int(input("Enter the first number: "))
b = int(input("Enter the second number: "))
lcm = calculate_lcm(a, b)
print(f"The LCM of {a} and {b} is {lcm}."
```

```
PS D:\AI assisted coding> & C:/Users/vaish/AppData/Local/Programs/Python/Python314/python.exe "d:/AI assisted coding/assignment_3.5.py"
Enter the first number: 4
Enter the second number: 6
The LCM of 4 and 6 is 12.

PS D:\AI assisted coding> & C:/Users/vaish/AppData/Local/Programs/Python/Python314/python.exe "d:/AI assisted coding/assignment_3.5.py"
Enter the first number: 5
Enter the second number: 10
The LCM of 5 and 10 is 10.

PS D:\AI assisted coding> 7
```

Question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function

that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010

Task:

- Compare clarity with zero-shot output.
- Analyze handling of zero and negative numbers.

The screenshot displays a VS Code workspace with three main components:

- Editor (assignment_3.5.py):** Contains a Python function `decimal_to_binary` that converts a decimal integer to a binary string. It includes a docstring with a return type and an example usage.
- Terminal:** Shows the execution of the script. It prompts for a decimal number (3), calculates its LCM (21), and then demonstrates the `decimal_to_binary` function with inputs 1010 and 10, showing the correct binary outputs.
- Chat Window:** Contains a one-shot prompt for writing a Python function named `decimal_to_binary` that accepts an integer and returns its binary representation as a string. It includes the example input/output and a task instruction to ensure correct handling of zero and negative numbers.

Question 6: Few-Shot Prompting (Harshad Number Check)

Write a few-shot prompt to generate a Python function that checks

whether a number is a Harshad (Niven) number.

Examples:

- Input: 18 → Output: Harshad Number
- Input: 21 → Output: Harshad Number
- Input: 19 → Output: Not a Harshad Number

Task:

- Test boundary conditions.
- Evaluate robustness

The screenshot shows an IDE with a Python file named `assignment_3.5.py`. The code defines a function `check_harshad_number(num: int) -> bool:` that checks if a number is a Harshad number. The function calculates the sum of its digits and checks if the number is divisible by that sum. It includes a docstring and an example usage section. The terminal output shows the function being tested with inputs 18, 21, and 19, resulting in '18 is a Harshad number.', '21 is a Harshad number.', and '19 is not a Harshad number.' respectively. The chat window on the right shows a few-shot prompt for the function.

```
def check_harshad_number(num: int) -> bool:
    """
    num (int): The integer to check.

    Returns:
        bool: True if the number is a Harshad number, False otherwise.
    """
    if num <= 0:
        return False  # Harshad numbers are typically positive integers
    digit_sum = sum(int(digit) for digit in str(num))
    return num % digit_sum == 0

# Example usage:
num = int(input("Enter a number: "))
if check_harshad_number(num):
    print(f"{num} is a Harshad number.")
else:
    print(f"{num} is not a Harshad number.")
```

Terminal Output:

```
PS D:\AI assisted coding> & C:/Users/vaish/AppData/Local/Programs/Python/Python314/python.exe "d:/AI assisted coding/assignment_3.5.py"
Enter a number: 18
18 is a Harshad number.
PS D:\AI assisted coding> & C:/Users/vaish/AppData/Local/Programs/Python/Python314/python.exe "d:/AI assisted coding/assignment_3.5.py"
Enter a number: 21
21 is a Harshad number.
PS D:\AI assisted coding> & C:/Users/vaish/AppData/Local/Programs/Python/Python314/python.exe "d:/AI assisted coding/assignment_3.5.py"
Enter a number: 19
19 is not a Harshad number.
PS D:\AI assisted coding>
```

Chat Prompt:

```
< num: int -> bool:
integer is a Harshad (Niven) number
an integer that is divisible by
integer to check.
the number is a Harshad number, False
Harshad numbers are typically positive integers
digit for digit in str(num)
sum == 0
num: int
num):
harshad number.)
a Harshad number.)
```